Mathematical Physics I (Fall 2026): Midterm Exam Solution

Oct. 24, 2026

[total 20 pts, closed book/cellphone, no calculator, 90 minutes]

1. (a) [1 pt] Find the first three terms of the Maclaurin series for the following function:

$$\int_0^x \frac{dt}{\sqrt{1+t^2}}.$$

(b) [2 pt] Consider a state of a system (e.g., a particle in a box, or a quantized oscillator) whose energy when occupied by a single particle is ϵ_0. In the quantum statistical distribution of bosons with unspecified total number of particles (so-called *Bose-Einstein* statistics), the expected total energy of all the particles in this state is found to be

$$\bar{\epsilon} = \frac{\sum_{n=1}^{\infty} n\epsilon_0 e^{-\frac{n\epsilon_0}{k_B T}}}{\sum_{n=0}^{\infty} e^{-\frac{n\epsilon_0}{k_B T}}}, \tag{1}$$

where k_B is the Boltzmann constant, and T is the temperature. Show that the ratio becomes

$$\bar{\epsilon} = \frac{\epsilon_0}{e^{\frac{\epsilon_0}{k_B T}} - 1}$$

by first identifying Eq.(1)'s denominator as a binomial expansion of $\frac{1}{1-x}$ with $x = e^{-\frac{\epsilon_0}{k_B T}}$, and its numerator as a constant times $\frac{x}{(1-x)^2}$. Using a power series expansion, show also that $\bar{\epsilon}$ reduces to $k_B T$ in the classical limit of $k_B T \gg \epsilon_0$.

• (a) From Eq.(13.5) of Boas Chapter 1, you can get the binomial series expansion $(1+t^2)^{-\frac{1}{2}} \simeq 1 + (-\frac{1}{2})t^2 + \frac{(-\frac{1}{2})(-\frac{3}{2})}{2!}t^4$. Integrating both sides, you find $\int_0^x (1+t^2)^{-\frac{1}{2}} dt \simeq x - \frac{1}{6}x^3 + \frac{3}{40}x^5$.

• (b) By differentiating $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, you find $\frac{x}{(1-x)^2} = \sum_{n=1}^{\infty} nx^n$, which leads to

$$\bar{\epsilon} = \frac{\sum_{n=1}^{\infty} \epsilon_0 n x^n}{\sum_{n=0}^{\infty} x^n} = \frac{\epsilon_0 x}{1-x} = \frac{\epsilon_0}{x^{-1}-1}.$$

Then, with $y = \frac{\epsilon_0}{k_B T} \ll 1$, you get $\bar{\epsilon} = \frac{\epsilon_0}{e^y - 1} \simeq \frac{\epsilon_0}{1+y-1} = k_B T$.

2. (a) [2 pt] Show that $\arcsin z = -i \ln (iz \pm \sqrt{1-z^2})$, where z is a complex number.

(b) [1 pt] Using the *principal value* of the complex logarithm, evaluate the expression $i^{\ln i}$ in simple $x + iy$ form.

• (a) $\arcsin z = \omega \;\rightarrow$ With $u \equiv e^{i\omega}$, you find $z = \sin\omega = \frac{e^{i\omega}-e^{-i\omega}}{2i} = \frac{u-u^{-1}}{2i} \;\rightarrow\; u^2 - 2izu - 1 = 0$ $\rightarrow\; \omega = -i \ln u = -i \ln(iz \pm \sqrt{1-z^2}) = \arcsin z$.

• (b) Using the definition of complex powers in Eq.(14.1) of Boas Chapter 2, $i^{\ln i} = e^{(\ln i)(\ln i)} = e^{\left\{i\left(\frac{\pi}{2}+2n\pi\right)\right\}\cdot\left\{i\left(\frac{\pi}{2}+2m\pi\right)\right\}} = e^{-\pi^2\left\{\frac{1}{4}+(n+m)+4nm\right\}}$. Then, taking the principal value (i.e., $n = m = 0$) yields $e^{-\frac{\pi^2}{4}}$.

3. (a) [2 pt] Solve the set of equations below by <u>*all three*</u> methods listed here: *(i)* by row reducing the augmented matrix, *(ii)* by using Cramer's rule, *(iii)* by finding the inverse of the coefficient matrix. Clearly indicate which method you are using for each part of your answer, so that the grader could follow it easily.

$$\begin{cases} 2x - \;\; y + \;\; z = 5 \\ \;\; x + 2y - \;\; z = 4 \\ 3x + \;\; y + 2z = 17 \end{cases}$$

(b) [2 pt] Verify that the following matrix H is Hermitian. Find its eigenvalues and eigenvectors, and write a unitary matrix U which diagonalizes H by a similarity transformation. Finally, show that $U^{-1}HU$ yields the diagonal matrix of eigenvalues.

$$H = \begin{pmatrix} -2 & 3+4i \\ 3-4i & -2 \end{pmatrix}$$

(c) [1 pt] Consider a particle moving in the xy-plane near the origin. For small displacements from the origin, suppose that the force acting on the particle is given by

$$\mathbf{F} = -\begin{pmatrix} 4 & 3 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = -M\mathbf{x}.$$

Notice that, in general, the x- and y-motion are *coupled* — e.g., a displacement along the x-axis does not produce a force purely in the x-direction. Now, consider a sufficiently small displacement $\mathbf{x} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$. Show that the force is parallel to the displacement only along certain special directions (so-called *principal* directions). Explain why this is an eigenvalue problem, and find all eigenvalues and their corresponding normalized eigenvectors. Finally, identify which principal direction experiences the stronger restoring force.

(d) [1 pt] Now consider a modified force law,

$$\mathbf{F} = -\begin{pmatrix} 4 & 5 \\ 5 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = -M'\mathbf{x}.$$

Find the eigenvalues and their normalized eigenvectors for this new matrix. Show that one of the eigenvalues is negative and the other is positive, and explain what this means physically for the stability of the equilibrium at the origin.[1]

[1] This behavior is characteristic of a *saddle point* discussed in Boas Chapter 4, Section 8.

• (a-1) As in Example 1 of Boas Chapter 3, Section 2,

$$\begin{pmatrix}2&-1&1&5\\1&2&-1&4\\3&1&2&17\end{pmatrix} \to \begin{pmatrix}1&2&-1&4\\2&-1&1&5\\3&1&2&17\end{pmatrix} \to \begin{pmatrix}1&2&-1&4\\0&-5&3&-3\\0&-5&5&5\end{pmatrix} \to \begin{pmatrix}1&2&-1&4\\0&-5&3&-3\\0&0&2&8\end{pmatrix}$$
$$\to \begin{pmatrix}1&2&-1&4\\0&1&-\frac{3}{5}&\frac{3}{5}\\0&0&1&4\end{pmatrix} \to \begin{pmatrix}1&2&-1&4\\0&1&0&3\\0&0&1&4\end{pmatrix} \to \begin{pmatrix}1&0&0&2\\0&1&0&3\\0&0&1&4\end{pmatrix}$$

• (a-2) As in Example 5 of Boas Chapter 3, Section 3,

$$x = \begin{vmatrix}4&2&-1\\5&-1&1\\17&1&2\end{vmatrix} \Big/ \begin{vmatrix}1&2&-1\\2&-1&1\\3&1&2\end{vmatrix} = \begin{vmatrix}4&2&-1\\9&1&0\\25&5&0\end{vmatrix} \Big/ \begin{vmatrix}1&2&-1\\0&-5&3\\0&-5&5\end{vmatrix} = -\begin{vmatrix}9&1\\25&5\end{vmatrix} \Big/ \begin{vmatrix}-5&3\\-5&5\end{vmatrix} = 2, \text{ etc.}$$

• (a-3) As in Example 3 of Boas Chapter 3, Section 6,

$$M = \begin{pmatrix}1&2&-1\\2&-1&1\\3&1&2\end{pmatrix} \to \det(M) = -10 \text{ and } C = \begin{pmatrix}-3&-1&5\\-5&5&5\\1&-3&-5\end{pmatrix}$$
$$\to M^{-1} = \frac{1}{\det(M)}C^T = \frac{1}{10}\begin{pmatrix}3&5&-1\\1&-5&3\\-5&-5&5\end{pmatrix} \to \text{ then, } x = 2, \text{ etc.}$$

• (b) From the characteristic equation, two eigenvalues $\lambda_1 = 3$ and $\lambda_2 = -7$ appear that correspond to eigenvectors $\mathbf{r}_1 = \frac{1}{5\sqrt{2}}\begin{pmatrix}5\\3-4i\end{pmatrix}$ and $\mathbf{r}_2 = \frac{1}{5\sqrt{2}}\begin{pmatrix}3+4i\\-5\end{pmatrix}$, respectively. Then it is straightforward to show that

$$U^{-1}HU = \frac{1}{50}\begin{pmatrix}5&3+4i\\3-4i&-5\end{pmatrix}\begin{pmatrix}-2&3+4i\\3-4i&-2\end{pmatrix}\begin{pmatrix}5&3+4i\\3-4i&-5\end{pmatrix} = \frac{1}{50}\begin{pmatrix}150&0\\0&-350\end{pmatrix} = \begin{pmatrix}\lambda_1&0\\0&\lambda_2\end{pmatrix}.$$

• (c) The principal directions satisfy $M\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ, which means that this is essentially an eigenvalue problem. From the characteristic equation, two eigenvalues $\lambda_1 = 7$ and $\lambda_2 = 1$ appear that correspond to eigenvectors $\mathbf{r}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$ and $\mathbf{r}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$, respectively. This means that a unit displacement along the direction $y = x$ gives the restoring force $\mathbf{F}_1 = -M\mathbf{r}_1 = -\lambda_1\mathbf{r}_1 = -7\mathbf{r}_1$, whereas a unit displacement along the direction $y = -x$ gives the restoring force $\mathbf{F}_2 = -M\mathbf{r}_2 = -\lambda_2\mathbf{r}_2 = -\mathbf{r}_2$.

• (d) From the characteristic equation, two eigenvalues $\lambda_1 = 9$ and $\lambda_2 = -1$ appear that correspond to eigenvectors $\mathbf{r}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$ and $\mathbf{r}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$, respectively. This means that a unit displacement along the direction $y = x$ gives the restoring force $\mathbf{F}_1 = -M'\mathbf{r}_1 = -\lambda_1\mathbf{r}_1 = -9\mathbf{r}_1$, whereas a unit displacement along the direction $y = -x$ gives the *repulsive* force $\mathbf{F}_2 = -M'\mathbf{r}_2 = -\lambda_2\mathbf{r}_2 = +\mathbf{r}_2$, away from the origin. Therefore, the equilibrium at the origin is stable along one principal direction but unstable along the other.

4. (a) [1 pt] Find the point on $2x + 3y + z - 12 = 0$ for which $2x^2 + y^2 + z^2$ is a minimum.

(b) [2 pt] If $x^2 + y^2 = 2st - 10$ and $2xy = s^2 - t^2$, find $\frac{\partial x}{\partial s}$, $\frac{\partial x}{\partial t}$, $\frac{\partial y}{\partial s}$, $\frac{\partial y}{\partial t}$ at $(x, y, s, t) = (4, 2, 5, 3)$.

• (a) From Thm.(9.14) of Boas Chapter 4, we write $F = 2x^2 + y^2 + z^2 + \lambda(2x + 3y + z)$. Then, $\frac{\partial F}{\partial x} = 4x + 2\lambda = 0$, $\frac{\partial F}{\partial y} = 2y + 3\lambda = 0$, and $\frac{\partial F}{\partial z} = 2z + \lambda = 0$. Combining the equations with the constraints $2x + 3y + z = 12$ gives $\lambda = -2$ and $(x, y, z) = (1, 3, 1)$.

• (b) Example 4 of Boas Chapter 4, Section 7 describes two ways to find the derivatives at one point when x and y are implicitly given as functions of s and t. Taking the second approach, we first find $8dx + 4dy = 6ds + 10dt$ and $4dx + 8dy = -6dt + 10ds$ at $(x, y, s, t) = (4, 2, 5, 3)$. Eliminating dy, we acquire $6dx = 13dt + ds$, which leads to $\frac{\partial x}{\partial s} = \frac{1}{6}$ and $\frac{\partial x}{\partial t} = \frac{13}{6}$.

5. [2 pt] Compute the gravitational force on a unit mass located at the origin, due to the mass of uniform density ρ occupying the volume inside the sphere $r = 2a$ (centered at the origin) and above the plane $z = a$. (Note: Make sure to specify both the magnitude and the direction of the resulting force. You may invoke symmetry arguments to explain why certain components of the net force are zero. The gravitational constant is G.)

• In spherical coordinates, the magnitude of the gravitational force on the unit mass due to the element of mass dM at (r, θ, ϕ) is $dF = \frac{GdM}{r^2} = \frac{G\rho dV}{r^2}$. So, the z-component of the force is

$$F_z = \int dF\cos\theta = \int \frac{G\rho}{r^2} \cdot r^2\sin\theta\, dr d\theta d\phi \cdot \cos\theta = G\rho \int_0^{2\pi} d\phi \int_0^{\frac{\pi}{3}} \sin\theta\cos\theta\, d\theta \int_{\frac{a}{\cos\theta}}^{2a} dr$$
$$= 2\pi aG\rho \int_0^{\frac{\pi}{3}} \sin\theta\cos\theta\, d\theta \left[2 - \frac{1}{\cos\theta}\right] = 2\pi aG\rho \int_0^{\frac{\pi}{3}} [\sin 2\theta - \sin\theta]\, d\theta = \frac{1}{2}\pi aG\rho,$$

where we adjusted the range of r as a function of θ, while keeping θ's range at $[0, \frac{\pi}{3}]$. Alternatively, you may adjust the range of θ as a function of r, while keeping r's range at $[a, 2a]$:

$$F_z = \int dF\cos\theta = \int \frac{G\rho}{r^2} \cdot r^2\sin\theta\, dr d\theta d\phi \cdot \cos\theta = G\rho \int_0^{2\pi} d\phi \int_a^{2a} dr \int_0^{\arccos\left(\frac{a}{r}\right)} \sin\theta\cos\theta d\theta$$
$$= 2\pi G\rho \int_a^{2a} dr \left[-\frac{1}{4}\cos 2\theta\right]_0^{\arccos\left(\frac{a}{r}\right)} = 2\pi G\rho \int_a^{2a} dr \left[-\frac{1}{4}(2\cos^2\theta - 1)\right]_0^{\arccos\left(\frac{a}{r}\right)}$$
$$= 2\pi G\rho \int_a^{2a} dr \left[\frac{1}{2}\left(1 - \frac{a^2}{r^2}\right)\right] = \pi G\rho \left[r + \frac{a^2}{r}\right]_a^{2a} = \frac{1}{2}\pi aG\rho.$$

Either way, you find that the force exerted by the "spherical cap" on the unit mass at the origin is $\mathbf{F} = F_z\hat{\mathbf{z}} = \frac{1}{2}\pi aG\rho\hat{\mathbf{z}}$.

6. (a) [1 pt] Prove that $(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) = (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C})$. Here you are asked to prove the identity by explicitly working with the components, i.e.,

$$(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) = \begin{vmatrix} (\mathbf{A} \times \mathbf{B})_x & (\mathbf{A} \times \mathbf{B})_y & (\mathbf{A} \times \mathbf{B})_z \\ C_x & C_y & C_z \\ D_x & D_y & D_z \end{vmatrix} = ...$$

You are welcomed to also tackle this problem using tensor notation for an additional +1 point.

(b) [1 pt] Evaluate the integral

$$\iint_\sigma (\nabla \times \mathbf{V}) \cdot \mathbf{n} d\sigma$$

over the entire surface σ of a pyramid whose base is the square in the (x, y)-plane with corners at $(0, 0)$, $(0, 2)$, $(2, 0)$, $(2, 2)$ and whose top vertex is at $(1, 1, 2)$, where $\mathbf{V} = (x^2 z - 2)\mathbf{i} + (x + y - z)\mathbf{j} - xyz\mathbf{k}$. Here, $\mathbf{n}$ denotes the unit normal vector to σ, pointing outward from the pyramid.

(c) [1 pt] Evaluate the integral

$$\iint_\sigma \mathbf{V} \cdot \mathbf{n} d\sigma$$

over the entire surface σ of the cylinder defined by $x^2 + y^2 = 16$, $z = 5$ and $z = -5$, where $\mathbf{V} = (y^2 - x^2)\mathbf{i} + (2xy - y)\mathbf{j} + 3z\mathbf{k}$. Here, $\mathbf{n}$ denotes the unit normal vector to σ, pointing outward from the cylinder.

• (a-1) From Eq.(3.2) of Boas Chapter 6,

$$\begin{aligned}
(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) &= \begin{vmatrix} (\mathbf{A} \times \mathbf{B})_x & (\mathbf{A} \times \mathbf{B})_y & (\mathbf{A} \times \mathbf{B})_z \\ C_x & C_y & C_z \\ D_x & D_y & D_z \end{vmatrix} \\
&= (A_y B_z - A_z B_y)(C_y D_z - C_z D_y) + (A_z B_x - A_x B_z)(C_z D_x - C_x D_z) \\
&\quad + (A_x B_y - A_y B_x)(C_x D_y - C_y D_x) \\
&= (A_x C_x + A_y C_y + A_z C_z)(B_x D_x + B_y D_y + B_z D_z) \\
&\quad - (A_x D_x + A_y D_y + A_z D_z)(B_x C_x + B_y C_y + B_z C_z) \\
&= (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C}).
\end{aligned}$$

• (a-2) Alternatively, writing in tensor notation and then using Eqs. (5.8) and (5.11) of Boas Chapter 10, $(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) = (\mathbf{A} \times \mathbf{B})_i (\mathbf{C} \times \mathbf{D})_i = \epsilon_{ijk} A_j B_k \cdot \epsilon_{imn} C_m D_n = (\delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}) A_j B_k C_m D_n = A_j B_k C_j D_k - A_j B_k C_k D_j = (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C})$.

• (b) Using the divergence theorem, Eq.(10.17) of Boas Chapter 6, and Eq.(d) in p.339 of Boas,

$$\iint_\sigma (\nabla \times \mathbf{V}) \cdot \mathbf{n} d\sigma = \oint \nabla \cdot (\nabla \times \mathbf{V})\, d\tau = 0$$

• (c) Again, using the divergence theorem, Eq.(10.17) of Boas Chapter 6,

$$\iint_\sigma \mathbf{V} \cdot \mathbf{n} d\sigma = \oint \nabla \cdot \mathbf{V} d\tau = \oint 2 d\tau = 2 \cdot 10 \cdot 16\pi = 320\pi$$